KETAN JOSHI

Software Developer | MCA Candidate
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PROFESSIONAL SUMMARY

Software developer pursuing MCA with demonstrated experience shipping full-stack web applications, AI-powered developer tools, and a Chrome extension used across 9 AI platforms. Proficient in Python, JavaScript, FastAPI, React, and PostgreSQL. Core competencies include multi-provider LLM orchestration, async API architecture, prompt engineering, OAuth/JWT authentication, and real-time WebSocket communication.

EDUCATION

Master of Computer Applications (MCA) | Ganpat University, Kherva, Gujarat | 2024 - 2026 (Pursuing)

Bachelor of Computer Applications (BCA) | Ganpat University, Kherva, Gujarat | Completed in 2024

TECHNICAL SKILLS

- Languages: Python, JavaScript, Java, SQL, HTML, CSS
- Frameworks & Libraries: FastAPI, React, Vite, Express, Flask, React Flow, Tailwind CSS, SQLAlchemy, Pydantic
- Databases: PostgreSQL (AsyncPG), MongoDB
- Tools & Platforms: Git/GitHub, Node.js, Chrome Extensions (Manifest V3), WebSocket, JWT, Google OAuth 2.0, Vercel
- Core Strengths: Full-stack application design, async API development, multi-provider LLM integration, prompt engineering, real-time systems

SELECTED PROJECTS

PromptOps | React 19, Vite, FastAPI, PostgreSQL, Gemini, OpenAI, Groq

- Designed and built a full-stack prompt engineering platform that transforms structured product requirements into deployment-ready prompts optimized for AI-native IDEs like Cursor, Antigravity, and Claude Dev, reducing AI iteration cycles.
- Engineered a modular prompt engine with role-based template composition, three AI control modes (Controlled, Balanced, Exploratory), dynamic tech-stack rule selection, and multi-provider LLM enhancement with automatic failover across Gemini 2.0 Flash, OpenAI GPT-4o, and Groq Llama.
- Implemented a visual chain execution system using React Flow with topological sort (Kahn's algorithm), cycle detection, and inter-step context passing for complex multi-stage prompt workflows.
- Built secure JWT + Google OAuth 2.0 authentication with OTP email verification, a tiered credit billing system, multi-session chat with model selection and auto-save, and an async middleware stack including rate limiting (SlowAPI), security headers (HSTS, CSP), and CORS.

Synapse | Chrome Extension (Manifest V3), JavaScript, WebSocket, Node.js

- Built a real-time AI-to-code sync tool comprising a Chrome extension and local WebSocket server that extracts code blocks from 9 AI chat platforms — Claude, ChatGPT, Gemini, DeepSeek, Copilot, Grok, Poe, Mistral, and HuggingFace Chat — and writes them directly into local project files.
- Implemented a 5-method filename detection pipeline (code fence attributes, first-line comments, DOM labels, conversation context, file tree matching) with three write modes: full overwrite, SEARCH/REPLACE diffs with fuzzy matching, and smart patch via existing-code markers.
- Developed an interactive extension popup with a file tree and content viewer, real-time connection status, operation statistics, and activity logging, with configurable git backup, dry-run preview, path blocking, and auto-reconnect using exponential backoff.

Digital Reading Room (DRR) | React, Vite, Express, MongoDB

- Developed a full-stack document management platform with JWT authentication, PDF upload and viewer, notes system, analytics dashboard with reporting routes, email notification workflows, OCR text extraction via Tesseract.js, and a scheduled cron service.

GUNI COMS2-2026 Conference Website | React, Vite, Tailwind CSS, Express, MongoDB

- Built a responsive conference website for the International Conference on Computing Communication Security with dynamic content pages (Home, About, Authors, Call for Papers, Contact), a MongoDB seed script for initial data, an admin panel for content management, and Vercel deployment.

ADDITIONAL PROJECTS

- Ketan Portfolio: Personal portfolio site with dynamic GitHub API integration to auto-load latest repositories, responsive layout, Typed.js typing effects, ScrollReveal.js animations, and FontAwesome icons.
- TrafficTelligence: ML-based traffic volume prediction app using Python, Flask, scikit-learn (trained model + label encoder), Jupyter Notebook for EDA, and an HTML form interface.
- NewsApp: React news aggregator consuming a third-party News API with category filtering, pagination, and responsive card layout — built in both class-based and function-based component versions.
- SRP (Student Result Prediction): Flask web app that predicts student exam scores from study-related inputs using a trained Pandas/scikit-learn pipeline.

CERTIFICATIONS

- Big Data Analytics Foundation Course - A+ Grade, Ganpat University (Skill India Program)
- IT-ITeS SSC NASSCOM - Experiential Learning Certificate
- Microsoft Get Started with AI Agent Component Certificate
- Kotlin Programming Certificate - Infosys Springboard
- Python Programming (Parts 1, 2 and 3) - Infosys Springboard
- Software Engineering Certificate - Infosys Springboard
- UML Training - Infosys Springboard

LANGUAGES

- English, Gujarati, Hindi